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INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

Industrial Performance and Economic Growth: A Statistical Analysis of Factory Output and Capital Investments

HS163 Project Report

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Rushil Lakdawala	230106058
Rehan Sherawat	230123077
Abhiraj Mandal	230107005
Raman Mishra	230103082
Advaith Giridhar	230108077
Manya Aggarwal	220123033
Gaurav Singh Shekhawat	230104040

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1 Introduction

The data is a comprehensive statistical account of industrial activity over several financial years in India. It includes a range of economic, financial, and operational criteria applicable to factory activity. Hence it is suitable for policy formulation. Framed in a time-series format, the raw data has distinct rows that denote financial years and have corresponding indicators that enable one to measure industrial performance, employee participation, capital spending, and production efficiency.

The data covers a long period and captures the key essence of industrial advancement, such as the number of operating factories, circulating and fixed capital, the number of personnel, wage distribution, expenses on capital, and overall production outputs. In addition, it includes indicators relevant to economic value generation, such as gross and net value added, which measure the overall contribution of the industrial sector to the local economy. Gross value added is calculated without accounting for depreciation, whereas net value includes deductions and depreciation.

One of the key advantages of this dataset is its capacity to document temporal trends. Through examination of yearly fluctuations, trends can be identified, indicating growth, stagnation, or decline of industries. Such analysis can reveal whether the number of factories has grown, whether capital investment has increased, and how these changes relate to workforce size and overall productivity. The data set also reveals capital formation, which is important for the understanding of fluctuations in investment in industries over time and the long-term impact on economic growth. Another significant feature of this dataset is its application in the analysis of employment trends. It not only captures the number of employees directly employed in factories but also includes other factors, such as total persons engaged and honorarium paid (included as emoluments). *However, despite this, in our regression analysis, we have ignored time for the sake of simplicity. Hence, the relationship between the value of output and other factors does NOT depend on time, for our purposes.*

Indicators like total inputs, gross output, and net value added allow for the determination of the degree to which factories are making efficient use of inputs and verify whether industrial production is creating substantial economic value. The addition of capital formation indicators, such as gross capital formation and net fixed capital formation shows long term investment patterns.

2 Data and Methodology

The dataset under investigation is derived from the National Data and Analytics Platform (NDAP), published by the Reserve Bank of India (RBI) for the 2023–2024 period. It comprises a comprehensive panel of industrial indicators recorded over multiple financial years for various states in India. Data includes investment parameters (e.g., fixed capital, working capital), labor-related variables (e.g., persons engaged in factories), and output measures (value of gross output).

There are some peculiarities in the dataset. In columns where we should expect only whole number values, such as 'Number of Persons Engaged in Factories', we see decimal point numbers. How do we explain this? The answer lies in the division of the State of Andhra Pradesh, in 2014, into **Andhra Pradesh and Telangana**. Complicated calculations based on area of territories may have led to this. We address these peculiarities using rounding off of the numbers involved. Moreover, in 2019, the erstwhile State of Jammu and Kashmir was split into the Union Territories of **Jammu and Kashmir and Ladakh**.

2.1 Economic Term Definitions and Variable Selection

A critical component of our analysis is grounded in the precise definitions and rationale for variable inclusion and exclusion.

- **ROWID:** A unique identifier for each row in the dataset.
- **Country:** The country in which the data is recorded (in our case, India).
- **State:** State name
- **LGD code:** A code used to identify the state or region (e.g., Jammu And Kashmir).
- **YearCode:** A coded representation of the year for the data entry
- **Year:** The actual year or financial period for the data entry (e.g., Financial Year 1991-1992).
- **Number of factories:** The total number of factories in the state or region during the given year.
- **Fixed capital:** The total value of physical assets (e.g., machinery, buildings) used for production in the factories.

- **Working capital:** The capital required for day-to-day operations, including inventory and short-term financial resources.
- **Physical working capital:** The value of physical resources (e.g., raw materials) used for ongoing production processes.
- **Productive capital:** The total value of capital available for productive use, including both fixed and working capital.
- **Invested capital:** The total financial investment in the factories, including both physical and financial capital.
- **Number of workers:** The total number of workers employed in the factories.
- **Persons engaged in factories:** The total number of people working in the factories, including both direct workers and other personnel involved in factory operations.
- **Total emoluments:** The total compensation (salaries, wages, etc.) paid to the workforce.
- **Total inputs:** The total value of all inputs used in the production process, including capital and labor.
- **Value of gross output:** The total value of goods and services produced by the factories during the year.
- **Net value added:** The value added to products or services after subtracting the cost of intermediate goods and services used in production.
- **Gross value added:** The total value added to products or services before subtracting the costs of intermediate goods and services.
- **Net fixed capital formation:** The net increase in fixed assets (after accounting for depreciation) over the year.
- **Gross fixed capital formation:** The total investment in fixed assets, including new investments in machinery, buildings, etc., without adjusting for depreciation.
- **Gross capital formation:** The total value of investments made in both fixed and working capital, indicating the overall investment in the economy or industry.

However, the raw data could not be directly used to perform our regression. Hence, we had to take some data cleaning measures which have been explained in the following sub-sections.

2.2 Data Cleaning

1. We drop the following columns from our dataset:

   (a) **ROW ID -** This column is an identifier for each row in the dataset. It doesn't contain meaningful data for prediction.

   (b) **State LGDcode -** This column provides integer labellings to each state. However, this isn't useful for our model because the states column already takes care of the same and hence, we can omit the need to create separate integer labels for them.

   (c) **Country -** All the values in this column are "India" and thus it has no significance to our regression task.

   (d) **Year Code -** This column represents the year in which the data was recorded. Since we want to analyze the value of gross output as a result of various other economic variables and thus we are not performing any time series analysis. As a result, this column has been dropped.

2. We also had to take care of the NULL values in our data. Through careful observation of the data we realised that such values occurred mostly for the three columns: Number Of Factories, fixed capital, working capital. Thus we used them to take care of the missing values. We verified that we had no NULL values in our dataset after this step.

3. Next, we had to bring uniformity in the datatype of the values in our columns. This is because some rows had floating point values for a lot of columns which should not have been the case. To deal with this, we round off the decimal values to their nearest integers.

2.3 Regression

To begin with our regression task, we conducted a correlation analysis between all the columns of the data and tried including all the independent variables in our first regression attempt. However it was obvious from the results that this was not ideal because of the extremely high VIF scores.

Hence, we conducted a socio-economic analysis of all the features in our dataset as a result of which we came across a lot of redundancies which have been explained below.

We employ heuristics, i.e., we try to reason why a term is worth including or excluding depending on its definition.

After this process we had the following six gestures preserved due to the following reasons:

- **Number of factories:** The "number of factories" is a direct indicator of the productive capacity of a region or industry. While columns like "Fixed capital" and "Invested capital" measure the financial resources invested in production, the number of factories essentially reflects the infrastructure available for generating output. More factories typically translate into more production units, so this column encapsulates the broader picture of production capacity.

- **Persons engaged in factories:** This variable represents the labor force within the production process. It covers aspects that could be linked to columns like "Number of workers" and "Working capital." While "Working capital" focuses on financial liquidity and operational flexibility, the number of persons engaged highlights the workforce involved in turning capital into productive output. The labor force is a critical determinant in generating gross output, which is why this column is preserved.

- **Total emoluments:** It represents the total compensation paid to employees, directly reflecting labor costs. Though other financial variables like "Net value added" or "Gross value added" touch on the economic value created, total emoluments are a more granular way of understanding the costs tied to labor, which can influence the output in terms of productivity and efficiency. By keeping this, we are accounting for how labor compensation might influence overall production.

- **Total inputs:** This variable is crucial because it aggregates the resources used in production (including materials, energy, and labor). It's a broader measure that can reflect a combination of variables like "Fixed capital" and "Invested capital," since these financial inputs are typically used to procure raw materials, machinery, and other production resources. Therefore, by retaining total inputs, we have indirectly captured the influence of other capital measures without keeping each one separately.

- **Net fixed capital formation:** This column captures investments in the economy, particularly in machinery and buildings that are meant to be used in the long term. It's a good proxy for understanding how capital-intensive an industry is and how this impacts the production capacity. While "Gross fixed capital formation" might seem relevant, "Net fixed capital formation" gives a more accurate picture of how much of the capital has been effectively utilized for production, considering depreciation. It preserves the key insights from gross fixed capital but removes the redundancy of non-adjusted capital formation.

- **Gross capital formation:** This feature is a broad measure of all capital investments, including both fixed and working capital. It provides an overarching view of economic investment into industries, capturing the essence of economic expansion and growth. While "Physical working capital" and "Productive capital" are more specific, this column encompasses the full spectrum of capital accumulation.

Hence, we performed the regression again with these 6 variables as our features, the result of which are shown below:

```
. regress valueofgrossoutput numberoffactories personsengagedinfactories totalemoluments totalinputs netfixedcapit
> alformation grosscapitalformation

      Source |       SS       df       MS              Number of obs =    1028
-------------+------------------------------           F(  6,  1021) = 5104.08
       Model |  6.7340e+17     6  1.1223e+17           Prob > F      =  0.0000
    Residual |  2.2451e+16  1021  2.1989e+13           R-squared     =  0.9677
-------------+------------------------------           Adj R-squared =  0.9675
       Total |  6.9585e+17  1027  6.7755e+14           Root MSE      =  4.7e+06

------------------------------------------------------------------------------
       valueofgrossoutput |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------------------------+---------------------------------------------------
        numberoffactories |   212.5107   87.85545     2.42   0.016     40.11279    384.9086
personsengagedinfactories |  -4.885777    1.49547    -3.27   0.001    -7.820322   -1.951231
          totalemoluments |    8.12913   .3524979    23.06   0.000     7.437427    8.820834
              totalinputs |   .4029123   .0162693    24.77   0.000     .3709871    .4348375
 netfixedcapitalformation |  -1.746176   .3983963    -4.38   0.000    -2.527945   -.9644073
    grosscapitalformation |    5.61299    .323012    17.38   0.000     4.979147    6.246833
                    _cons |   59238.78     188243     0.31   0.753    -310148.7    428626.2
------------------------------------------------------------------------------

. vif

    Variable |       VIF       1/VIF
-------------+----------------------
personseng~s |     23.79    0.042034
numberoffa~s |     18.86    0.053012
grosscapit~n |     11.64    0.085901
totalemolu~s |      9.11    0.109775
 totalinputs |      6.57    0.152159
netfixedca~n |      3.26    0.306698
-------------+----------------------
    Mean VIF |     12.21
```

Figure 1

It is evident that we still have some correlations left in our features, and based upon the high VIF scores we can easily observe that the following two features can be dropped:

- **Number of factories:** Even though it plays a vital role in gross value, the actual underlying is the number of people working in the factories and hence it is redundant to the feature "Persons engaged in factories".
- **Net fixed capital formation:** Gross capital formation provides a more comprehensive view of economic investment by encompassing both long-term fixed assets and short-term working capital, which are crucial for ongoing production. It reflects the total volume of investments entering the economy, including essential operational resources like raw materials and inventories, thus offering a dynamic and holistic picture of economic activity.

Thus, we reach our final regression model with the following results:

```
. regress valueofgrossoutput personsengagedinfactories totalemoluments totalinputs grosscapitalformation

      Source |       SS       df       MS              Number of obs =    1029
-------------+------------------------------           F(  4,  1024) = 7494.59
       Model |  6.7302e+17     4  1.6826e+17           Prob > F      =  0.0000
    Residual |  2.2989e+16  1024  2.2450e+13           R-squared     =  0.9670
-------------+------------------------------           Adj R-squared =  0.9668
       Total |  6.9601e+17  1028  6.7705e+14           Root MSE      =  4.7e+06

------------------------------------------------------------------------------
       valueofgrossoutput |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------------------------+---------------------------------------------------
personsengagedinfactories |  -1.739289   .5820021    -2.99   0.003    -2.881342   -.5972362
          totalemoluments |   8.557576   .3307408    25.87   0.000     7.908568    9.206583
              totalinputs |   .4100053    .016356    25.07   0.000     .3779102    .4421004
    grosscapitalformation |   4.616962   .2282691    20.23   0.000     4.169034    5.064891
                    _cons |   86777.83   189976.7     0.46   0.648    -286010.2    459565.9
------------------------------------------------------------------------------

. vif

    Variable |       VIF       1/VIF
-------------+----------------------
totalemolu~s |      7.86    0.127281
 totalinputs |      6.51    0.153677
grosscapit~n |      5.70    0.175566
personseng~s |      3.53    0.283208
-------------+----------------------
    Mean VIF |      5.90
```

Figure 2

Hence, our final model has four regressors: "Persons Engaged in Factories", "Total Emoluments", "Total Inputs", "Gross Capital Formation" and the regressand is the Value of Gross Output.

3 Discussion and Conclusion

- A crucial observation (among many) underlining the deviation of our model from an ideal regression model is the following: If we include the number of workers (hereafter in this section referred to as 'A') and number of people engaged in factories (hereafter in this section referred to as 'B') in our regression model, we would obtain coefficients of A and B as having opposite signs. However, when taken individually, i.e., exclusively A or B, these coefficients had the same signs. How do we explain this? The reason for this is the exceptionally high multicollinearity between these two variables: 0.9968. Thus, the coefficient estimates' opposite signs (in the case of A and B considered together) occur due to their high correlation; this distorts the true effect which they have in isolation.

```
. correlate numberoffactories fixedcapital workingcapital physicalworkingcapital productivecapital investedcapital numberofworkers personsengagedinfactories to
> talemoluments totalinputs valueofgrossoutput netvalueadded grossvalueadded netfixedcapitalformation grossfixedcapitalformation grosscapitalformation
(obs=1011)
```

	numbe~es	fixedc~l	workin~l	physic~l	produc~l	invest~l	numbe~rs	person~s	totale~s	totali~s	valueo~t	netval~d	grossv~d	netfix~n	grossf~n	grossc~n
numberoffa~s	1.0000															
fixedcapital	0.7104	1.0000														
workingcap~l	0.7315	0.7678	1.0000													
physicalwo~l	0.7944	0.9307	0.9127	1.0000												
productive~l	0.7461	0.9880	0.8571	0.9675	1.0000											
investedca~l	0.7460	0.9940	0.8224	0.9649	0.9965	1.0000										
numberofwo~s	0.9745	0.7327	0.7648	0.8211	0.7721	0.7700	1.0000									
personseng~s	0.9720	0.7394	0.7737	0.8292	0.7795	0.7771	0.9968	1.0000								
totalemolu~s	0.7874	0.8659	0.9241	0.9629	0.9184	0.9077	0.8295	0.8354	1.0000							
totalinputs	0.7186	0.8822	0.8463	0.9401	0.9126	0.9125	0.7535	0.7603	0.8997	1.0000						
valueofgro~t	0.7652	0.9320	0.8993	0.9912	0.9654	0.9634	0.7982	0.8062	0.9526	0.9508	1.0000					
netvaluead~d	0.7709	0.9014	0.9122	0.9740	0.9438	0.9364	0.8058	0.8196	0.9661	0.9171	0.9747	1.0000				
grossvalue~d	0.7770	0.9152	0.9101	0.9787	0.9544	0.9477	0.8111	0.8238	0.9680	0.9219	0.9788	0.9981	1.0000			
netfixedca~n	0.5420	0.6161	0.5082	0.6039	0.6165	0.6216	0.5438	0.5555	0.5375	0.5795	0.6131	0.6055	0.6080	1.0000		
grossfixed~n	0.7484	0.8888	0.7762	0.8900	0.9005	0.9023	0.7655	0.7757	0.8384	0.8448	0.8930	0.8824	0.8892	0.8888	1.0000	
grosscapit~n	0.7586	0.9021	0.8351	0.9373	0.9255	0.9260	0.7846	0.7949	0.8774	0.8800	0.9299	0.9206	0.9261	0.7743	0.9528	1.0000

Figure 3: Correlation Matrix showing the correlations between pairs of variables

- Now, we check for heteroscedasticity in our data. We get a residual v/s fitted plot as follows (Figure 4). The unequal spread of residuals suggests the presence of heteroscedasticity. This violates the classical linear regression assumption that residuals should be homoscedastic (i.e., have constant variance across all levels of the fitted values).

  Our *hettest* returned the following result as shown in Figure 5.

  Null hypothesis: Homoscedasticity holds.

  Conclusion from results: p-value is very low hence the null hypothesis is rejected. Therefore, we take measures to deal with heteroscedasticity in our data by performing regression along with robust command. Hence, we regress our model with robust commands.

- **Analysing serial correlation:** We clustered our regression model over the state variable and obtained the following result as in Figure 6.

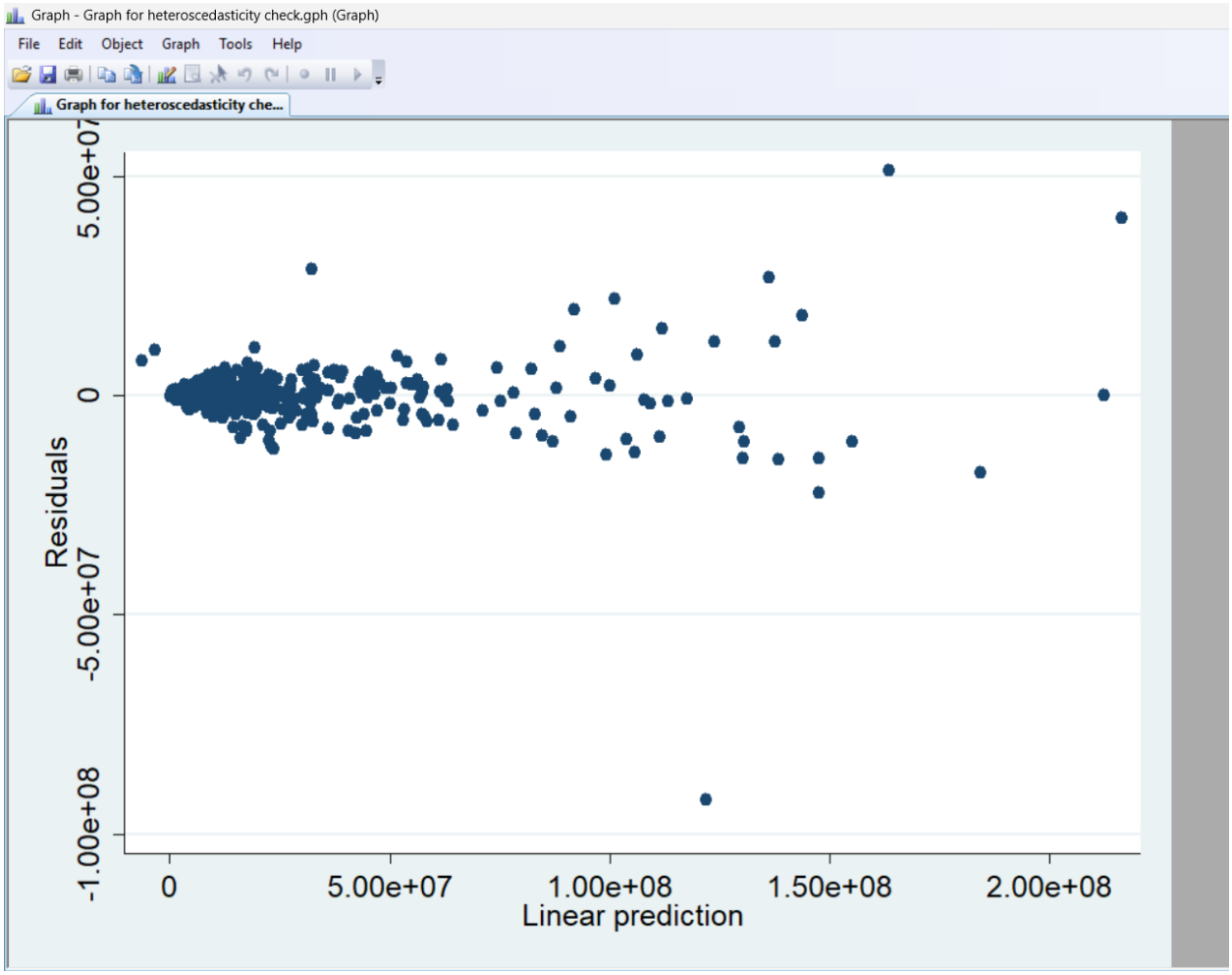

Figure 4

```
. hettest

Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
         Ho: Constant variance
         Variables: fitted values of valueofgrossoutput

         chi2(1)      =  7992.94
         Prob > chi2  =   0.0000
```

Figure 5

```
. regress valueofgrossoutput numberofworkers totalinputs totalemoluments grosscapitalformation, robust

Linear regression                                      Number of obs =    1029
                                                       F(  4,  1024) =  935.38
                                                       Prob > F      =  0.0000
                                                       R-squared     =  0.9670
                                                       Root MSE      =  4.7e+06

------------------------------------------------------------------------------
             |               Robust
valueofgrossoutput |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
numberofworkers | -2.263259   1.476806    -1.53   0.126    -5.161171    .6346528
totalinputs |  .4101032   .2584585     1.59   0.113    -.0970657    .9172722
totalemoluments |  8.568513   2.720793     3.15   0.002     3.229546    13.90748
grosscapitalformation |  4.605267   1.699224     2.71   0.007     1.270908    7.939625
       _cons |  93252.46   139119.6     0.67   0.503    -179739.6    366244.6
------------------------------------------------------------------------------
```

Figure 6

This shows an increase in the p values of the features as compared to our non-clustered regression. Hence we conclude that the features "Persons engaged in factories" and "Total emoluments" no longer stay significant. This indicates the presence of serial correlation in our data across the different states.

- **Takeaways:**

  1. Wages (total emoluments) have the strongest positive effect on output. A 1 standard deviation increase in wages increases output by 107 lakh rupees.
  2. Capital investment (gross capital formation) also significantly boosts output (71 lakh rupees per 1 standard deviation increase).
  3. Raw materials (total inputs) positively impact output, but slightly less than wages.
  4. Increasing the persons engaged reduces output. This suggests diminishing productivity, where too many workers without enough capital or resources create inefficiencies.
  5. A shift from the 25th to the 75th percentile in wages (total emoluments) increases output by 47.4 lakh rupees, reinforcing the importance of well-paid workers.
  6. Increasing capital investment (gross capital formation) in the same range raises output by 42.6 lakh units, making it the second most influential factor.
  7. Raw materials (total inputs) also increase output but to a slightly lesser extent.
  8. Hiring more workers from the 25th to the 75th percentile actually reduces output by 896,796 rupees. This suggests labor inefficiency and the need for better capital allocation instead of hiring more people.

9. Raising worker wages (total emoluments) from the mean to the 90th percentile increases output by 75.3 lakh rupees, emphasizing labor productivity over quantity.
10. Increasing capital investment (gross capital formation) has a strong effect of 66 lakh rupees.
11. Expanding raw material use (total inputs) also improves output but less than wages.
12. Hiring more workers reduces output further, reinforcing the need for capital-intensive production instead of labor expansion.

4 Policy Implications

Factories should improve wages while hiring lower-paid workers. This increases the gross value of output. Wages and then capital investment deliver the highest increase in production. Firms should invest in wages and capital more than in expanding the workforce that they employ. This means that a capital-intensive model is better than a labour-intensive model in terms of production.